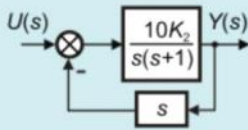


Model examen

Thursday, February 4, 2021

9:14 AM

Se consideră schema bloc a unui proces ca în figura de mai jos:



a. Să se determine funcția de transfer a procesului și să se construiască schema bloc de reglare

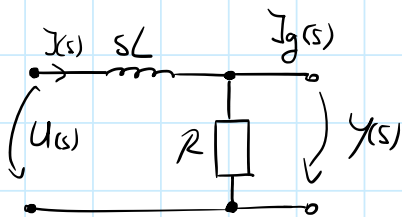
($G_R(s)=K_1$, $G_{EE}(s)=1$ și $G_{TR}(s)=1$); (1p)

b. Să se determine K_1 și K_2 , știind că $K_V=10$ și $\zeta=0.5$; (1p)

c. Să se calculeze t_v , M_v și t_s să se schițeze răspunsul sistemului la o mărime de intrare treaptă unitară (pe grafic se va preciza suprareglajul); (1p)

d. În sistem apare o perturbăție treaptă unitară la mărimea de comandă. Refaceți schema bloc a sistemului și determinați eroarea staționară pentru o mărime de intrare perturbăție treaptă $51_+(t)$. (1p)

e. Să se adauge un zero astfel încât performanțele sistemului să nu se modifice. (0.5p)



$$\frac{Y(s)}{U(s)} = \frac{I(s)R}{I(s)(R+sL)} \Rightarrow \frac{Y(s)}{U(s)} = \frac{R}{R+sL}$$

$$Y(s)(R+sL) = U(s) \cdot R \quad | \mathcal{L}^{-1} \Rightarrow R y(t) + \frac{dy(t)}{dt} \cdot L = R u(t)$$

$$y(0) = a$$

$$u(t) = 1_+(t)$$

$$\mathcal{L}\{R u(t)\} = R \cdot \frac{1}{s} \quad ; \quad \mathcal{L}\{R y(t)\} = R \cdot Y(s)$$

$$\mathcal{L}\left\{\frac{dy(t)}{dt} \cdot L\right\} = L(sY(s) - a) = L \cdot sY(s) - La$$

$$R \cdot u(s) + L \cdot s \cdot Y(s) - La = R \cdot \frac{1}{s}$$

$$Y(s)(R+Ls) = \frac{R}{s} + La \quad | :s$$

$$Y(s) \cdot s(R+Ls) = R + La \cdot s \Rightarrow Y(s) = \frac{R+La \cdot s}{R+Ls}$$

$$y(s) \cdot s(R+Ls) = R + La \cdot s \Rightarrow y(s) = \frac{R + La \cdot s}{s(R+Ls)}$$

$$y(s) = \frac{R}{s(R+Ls)} + \frac{La \cdot s}{s(R+Ls)}$$

$$y_p(s) = \frac{R}{s(R+Ls)} = \frac{C_1}{s} + \frac{C_2}{R+Ls}$$

$$C_1 = s y(s) \Rightarrow C_1 = \frac{R}{R+Ls} \Big|_{s=0} \Rightarrow C_1 = 1$$

$$C_2 = (R+Ls) y(s) \Rightarrow C_2 = \frac{R}{s} \Big|_{s=-\frac{R}{L}} \Rightarrow C_2 = -L$$

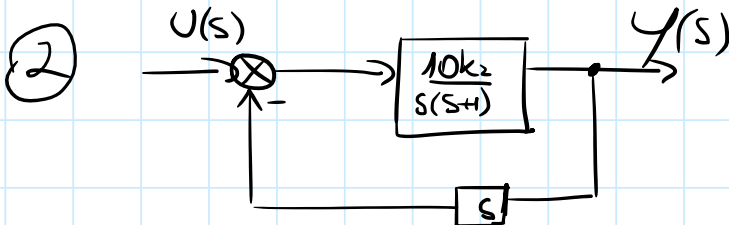
$$y_p(t) = 1_+(t) - \frac{L}{Ls+R} \rightarrow y_1(t) = 1_+(t) - L \cdot \frac{1}{L(s+R/L)}$$

$$y_1(t) = 1_+(t) - \frac{1}{s+R/L} \Rightarrow y(t) = \underline{1_+(t) - e^{-\frac{R}{L}t}} \rightarrow \text{răsp forat}$$

$$y_u(s) = \frac{L \cdot a \cdot s}{s(R+Ls)} = \frac{La}{R+Ls}$$

$$y_u(s) = ka \cdot \frac{1}{\left(\frac{R}{L} + s\right)} = a \cdot e^{-\frac{R}{L}t} \rightarrow \text{răsp natural}$$

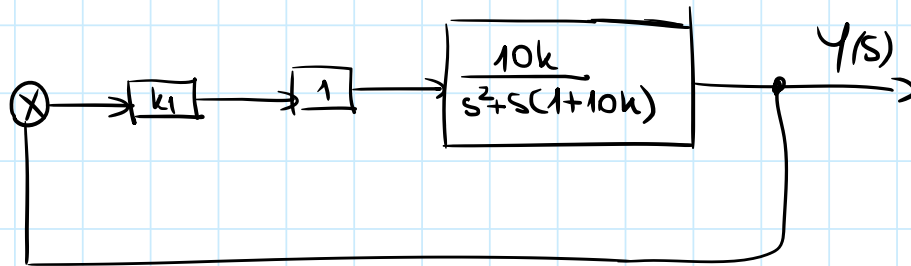
$$y(t) = y_u(t) + y_1(t) = 1_+(t) - e^{-\frac{R}{L}t} + a \cdot e^{-\frac{R}{L}t}$$



A) det funct de transfer stiind ca $GR(s) = k_1$, $GEE(s) = 1$ si $GTR(s) = 1$

$$G_0(s) = \frac{\frac{10k_2}{s(s+1)}}{1 + \frac{10k_2}{s(s+1)}} = \frac{\frac{10k_2}{s(s+1)}}{\frac{s(s+1) + 10k_2}{s(s+1)}} = \frac{10k_2}{s^2 + s + 10k_2} = \frac{10k_2}{s^2 + s(1+10k_2)}$$

$$G_0(s) = \frac{s(s+1)}{1 + \frac{10k_2}{s(s+1)}} = \frac{\cancel{s(s+1)}}{\frac{s(s+1) + 10k_2}{\cancel{s(s+1)}}} = \frac{10k_2}{s^2 + s + 10k_2s} = \frac{10k_2}{s^2 + s(1+10k_2)}$$



b) det k_1 & k_2 sind cā $k_v = 10$ & $\zeta = 0,5$

$$G_0(s) = \frac{10k_2 \cdot k_1}{s^2 + s(1+10k_2)} \quad \text{-- sis. desclis}$$

$$k_v = 10 \Rightarrow \lim_{s \rightarrow 0} s \cdot \frac{10k_2 \cdot k_1}{s^2 + s(1+10k_2)} = 10 \Rightarrow \lim_{s \rightarrow 0} \frac{10k_1 \cdot k_2}{s + 1 + 10k_2} = 10$$

$$\Rightarrow \frac{10k_1 \cdot k_2}{1 + 10k_2} = 10$$

$$G = \frac{\frac{10k_1 k_2}{s^2 + s(1+10k_2)}}{\frac{s^2 + s(1+10k_1 k_2) + 10k_1 k_2}{s^2 + s(1+10k_2)}} = \frac{10k_1 k_2}{s^2 + s(1+10k_1 k_2) + 10k_1 k_2} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = 10k_1 \cdot k_2$$

$$\begin{matrix} * & 1+10k_2 = 2\zeta\omega_n \\ & \zeta = 0,5 \end{matrix} \quad \Bigg| \Rightarrow \quad 1+10k_2 = \omega_n \Rightarrow 1+10k_2 = \sqrt{10k_2 \cdot k_1}$$

$$\frac{10k_2 \cdot k_1}{1+10k_2} = 10 \Rightarrow 10k_1 k_2 = 10 + 100k_2 \Rightarrow 0 = 10 + k_2(100 - k_1)$$

$$\Rightarrow k_2 = -\frac{10}{100 - k_1} = -\frac{1}{10 - k_1} \quad *$$

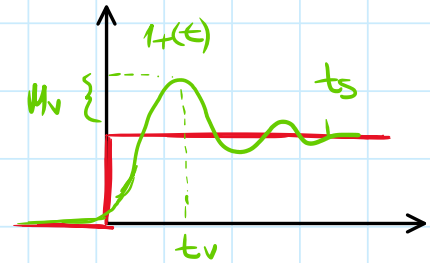
$$1 + 10 \left(\frac{-1}{10 - k_1} \right) = \sqrt{10 \cdot \frac{1}{10 - k_1} \cdot k_1} \Rightarrow 1 - \frac{10}{10 - k_1} = \sqrt{\frac{-10 k_1}{10 - k_1}}$$

$$k_1 = \frac{100}{9}$$

$$k_2 = \frac{-1}{10 - k_1} = - \frac{1}{\frac{90 - 100}{9}} = \frac{9}{10}$$

$$\omega_n^2 = 10 \cdot \frac{9}{10} \cdot \frac{100}{9} = 100 \Rightarrow \omega_n = 10$$

c) t_v, t_s, M_v ? graphic



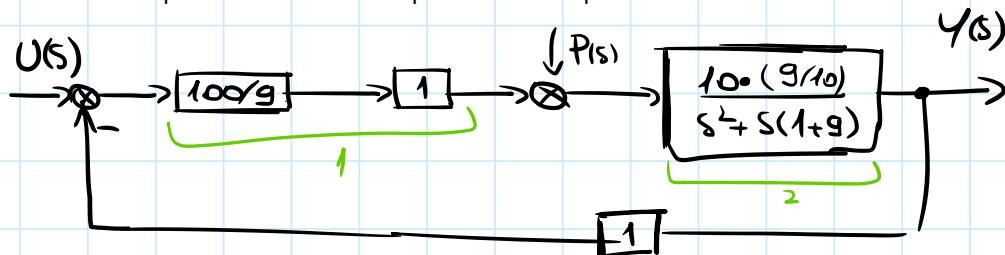
$$t_v = \frac{\pi}{\omega_n \sqrt{1 - \xi^2}} = \frac{\pi}{10 \sqrt{1 - 0.5^2}} = 0.36$$

$$M_v = e^{-\frac{\xi \pi}{\sqrt{1 - \xi^2}}} \Rightarrow M_v = e^{-\frac{0.5 \cdot \pi}{\sqrt{1 - 0.5^2}}} \Rightarrow M_v = 0.16$$

$$t_s = \frac{4}{\xi \omega_n} = \frac{4}{0.5 \cdot 10} = 0.8$$

d)

In schema apare o perturbatie treapta unitara la marimea de comanda. Refaceti schema bloc si deti eroarea stationala pt o marime de intrare perturbatie treapa



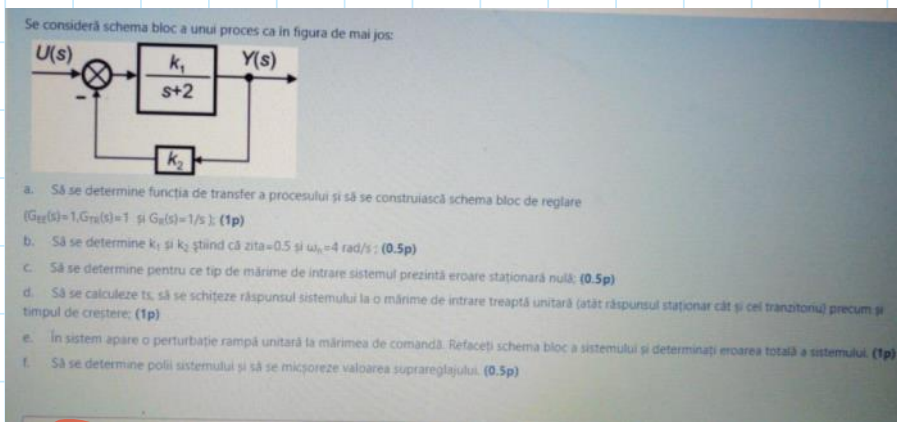
$$G_1 = \frac{100}{g} ; G_2 = \frac{9}{s^2 + 10s}$$

$$-5 \cdot \frac{A_1 \cdot B_2}{A_1 \cdot A_2 + B_1 \cdot B_2} = -5 \cdot \frac{9 \cdot 9}{90 + 100 \cdot 9} = \frac{9}{100} \cdot (-5) = -45/100$$

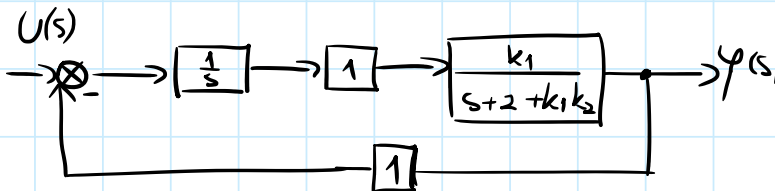
$$e) \frac{10 \cdot \frac{9}{10} \cdot \frac{100}{9}}{s^2 + s(1 + 10 \cdot \frac{9}{10}) + 10 \cdot \frac{9}{10} \cdot \frac{100}{9}} = \frac{100}{s^2 + 10s + 100}$$

adaug zero = -600

$$\frac{100(s+600)}{600s^2 + 600s + 600 \cdot 100}$$



a) $G_p = \frac{\frac{k_1}{s+2}}{1 + \frac{k_1}{s+2} \cdot k_2} = \frac{k_1}{s+2} \cdot \frac{s+2}{s+2+k_1k_2} = \frac{1}{s+2+k_1k_2}$



b) $G_e = \frac{\frac{1}{s} \cdot 1 \cdot \frac{k_1}{s+2+k_1k_2}}{\frac{1}{s} \cdot \frac{k_1}{s+2+k_1k_2} + 1} = \frac{\frac{k_1}{s^2+2s+k_1k_2}}{\frac{k_1}{s^2+2s+k_1k_2} + \frac{s^2+2s+k_1k_2}{s^2+2s+k_1k_2} + k_1}$

$$= \frac{k_1}{s^2+2s+k_1k_2} \cdot \frac{s^2+2s+k_1k_2}{k_1+s^2+2s+k_1k_2 \cdot s}$$

$$\frac{k_1}{k_1+s^2+2s+k_1k_2 \cdot s} = \frac{k_1}{s^2+2\zeta\omega_n s + \omega_n^2}$$

$$\frac{k_1}{k_1 s^2 + 2s + k_1 k_2} = \frac{k_1}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$\hookrightarrow \frac{k_1}{s^2 + s(2 + k_1 k_2) + k_1}$$

$$\Rightarrow \omega_n^2 = k_1 \mid \Rightarrow 4^2 = k_1 \Rightarrow k_1 = 16$$

$$2 + 16k_2 = 2 \cdot 0,5$$

$$2 + 16k_2 = 4 \mid \Rightarrow 16k_2 = 2 \Rightarrow k_2 = \frac{1}{8}$$

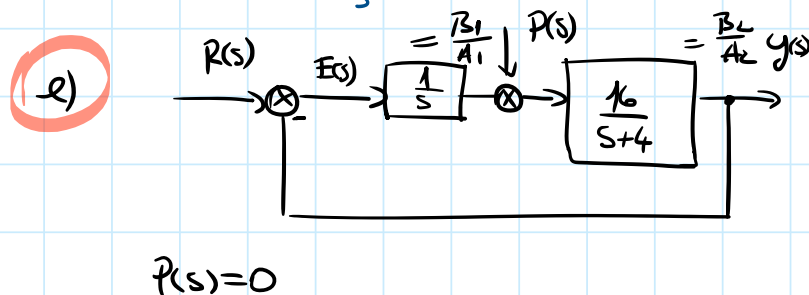
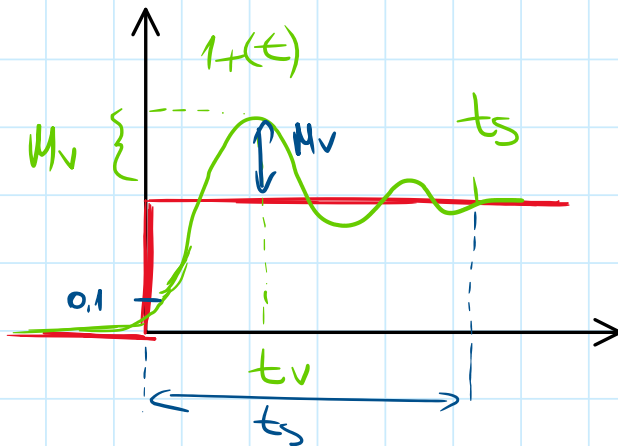
$$c) e_{st} = \frac{1}{1 + \lim_{s \rightarrow 0} (gd(s))} = \frac{1}{1 + \lim_{s \rightarrow 0} \frac{16}{s^2 + 4s}} = 0$$

ERAOARE STATIONALA NULA PT TREAPTA

$$d) t_c = \frac{\pi - \varphi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$\varphi = \arctg \frac{\sqrt{1 - \zeta^2}}{\zeta} = \arctg \frac{\sqrt{1 - 0,25}}{0,5} = \frac{0,36}{0,5} = 1,42$$

$$t_s = \frac{3}{0,5 \cdot 4} = 1,5 (s)$$



$$R(s) = \frac{1}{s} ; e_{st}' = \frac{1}{1 + \lim_{s \rightarrow 0} \frac{16}{s^2 + 4s}} = 0$$

$$= - \lim_{s \rightarrow 0} \frac{A_1(s) \cdot B_2(s)}{A_1(s) \cdot A_2(s) + B_1(s) \cdot B_2(s)}$$

$$= - \frac{A_1(0) \cdot B_2(0)}{A_1(0) \cdot A_2(0) + B_1(0) \cdot B_2(0)} = 0$$

$$e_{st}^+ = e_{st}' + e_{st}^p = 0 + 0 = 0$$

7) $\approx G_d$ (deschis)

Ex:

$$G_R \cdot G_{EE} \cdot G_P = \frac{1}{s} \cdot \frac{\frac{16}{s+2}}{s^2 + 1 + \frac{16}{s+2} \cdot \frac{1}{8}} = \frac{1}{s} \cdot \frac{16}{s+2} \cdot \frac{s+2}{s+4} = \frac{16}{s^2 + 4s}$$

$$s^2 + 4s = 0 \Rightarrow s(s+4) = 0 \Rightarrow \begin{matrix} p_1 = 0 \\ p_2 = -4 \end{matrix}$$

Microare Mv

$\Rightarrow$ adăugăm p_d „-2”

$$\Rightarrow G_o(s) = \frac{16}{s^2 + 4s(-\frac{s}{-2} + 1)} = \frac{16}{s^2 - 2s^2 + 4s} = \frac{16}{-s^2 + 4s}$$